

AIMS
Airbus Material Specification

Aluminium Alloy (7010/7050)
Solution treated controlled stretched
and artificially aged (T7651)

Plate

$6,0 \text{ mm} \leq a \leq 200,0 \text{ mm}$
Close tolerance flatness

Material Specification

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1 Scope

This Material Specification defines the requirements of aluminium alloy 7010/7050 T7651 plate $6,0\text{ mm} \leq a \leq 200,0\text{ mm}$ for aerospace application.
Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

No typical applications.

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ABS 5323	Aerospace series, Plates, Aluminium alloy 7010/7050, Close-tolerance flatness, Thickness $6,0\text{mm} \leq a \leq 200,0\text{ mm}$ Dimensions
AIMS 03-02-000	Technical specification – Aluminium and aluminium alloy wrought products – Plate
EN2032-2	Aerospace series – Metallic materials Part2 Coding of the metallurgical condition in the delivery condition.
EN4500-2	Aerospace series: Metallic materials: Rules for drafting and presentation of material standards Part 2:Specific rulesfor aluminium, aluminium alloys and magnesium alloys.
EN6072	Aerospace series – Metallic materials test methods constant amplitude fatigue testing
ASTM G34	Standard Test Method of exfoliation corrosion susceptibility in 2XXX and 7XXX series aluminium alloys (EXCO test)

4 Technical requirements

4.1 General

Specification values are specification minimums except where stated

4.2 Specific

The data requirements given in the tables are in accordance with the relevant material part of EN4500-2

5 Designation

Refer to ABS 5323

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS 03-02-000.

1	Material designation			Aluminium alloy 7010/7050 T7651										
2	Chemical Composition %	Element		Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Others		Al
		min.		See relevant IPS										
		max.												
3	Method of melting			-										
4	Form Method of production Limit dimension (mm)			Plate Rolled $6 \leq a \leq 200$										
5	Technical specification			AIMS 03-02-000										
6	6.1 Delivery condition and heat treatment			T7651 Solution treated $465 \leq \theta \leq 485^{\circ}\text{C}$ WQ $\theta \leq 40^{\circ}\text{C}$ Controlled stretched 1,5 – 3% Aged $117 \leq \theta \leq 123^{\circ}\text{C}/t=10\text{h}$ + $169 \leq \theta \leq 175^{\circ}\text{C}/t=8\text{h}$										
	6.2 Delivery condition code			U ¹⁾										
7	Use condition and heat treatment			T7651 As delivered										
8	Sample Test piece Heat treatment			T7651										
9	Dimension concerned	a	mm	$6 \leq a \leq 10$		$10 < a \leq 25$		$25 < a \leq 40$			$40 < a \leq 60$			
10	Thickness of cladding on each face		%	-										
11	Direction of test piece			L	LT	L	LT	L	LT	ST	L	LT	ST	
12	TENSILE	Temperature	θ	°C	Room temperature									
13		Proof stress	R _{p0,2}	MPa	450	450	450	450	450	450	420	450	445	420
14		Strength	R _m	MPa	525	525	525	525	520	520	500	520	520	500
15		Elongation	A _{5D}	%	8 ⁽²⁾	8 ⁽²⁾	8	6	8	6	4	8	6	4
16		Reduction of area	Z	%	-	-	-	-	-	-	-	-	-	-
17	Hardness		HB	-	-									
18	Shear strength		R _c	MPa	-									
19	Bending		k	-	-									
20	Impact strength		k	-	-									
21	CREEP	Temperature	θ	°C	-									
22		Time		h	-									
23		Stress	σ_a	MPa	-									
24		Elongation	a	%	-									
25		Rupture stress	σ_R	MPa	-									
26		Elongation at rupture	A	%	-									
27	Notes (see Line 98)			1), 2), 3), and 4)										

1	Material designation			Aluminium alloy 7010/7050 T7651											
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												Each	total		
		min.		See relevant IPS											
max.															
3	Method of melting			-											
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	6.2 Delivery condition code			U ¹⁾											
7	Use condition and heat treatment			T7651 As delivered											
8	Sample Test piece Heat treatment			T7651											
9	Dimension concerned	a	mm	$60 < a \leq 80$			$80 < a \leq 100$			$100 < a \leq 120$					
10	Thickness of cladding on each face		%	-											
11	Direction of test piece			L	LT	ST	L	LT	ST	L	LT	ST			
12	TENSILE	Temperature	θ	$^{\circ}\text{C}$	Room temperature										
13		Proof stress	R _{p0,2}	MPa	445	440	420	440	435	410	435	430	405		
14		Strength	R _m	MPa	505	510	500	500	505	480	495	500	475		
15		Elongation	A _{5D}	%	8	6	4	6	5	3	6	5	3		
16		Reduction of area	Z	%	-	-	-	-	-	-	-	-	-		
17	Hardness		HB	-	-										
18	Shear strength		R _c	MPa	-										
19	Bending		k	-	-										
20	Impact strength		k	-	-										
21	CREEP	Temperature	θ	$^{\circ}\text{C}$	-										
22		Time		h	-										
23		Stress	σ_a	MPa	-										
24		Elongation	a	%	-										
25		Rupture stress	σ_R	MPa	-										
26		Elongation at rupture	A	%	-										
27	Notes (see Line 98)			1), 2), 3), and 4)											

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		min.		See relevant IPS										
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	6.2 Delivery condition code			U ¹⁾										
7	Use condition and heat treatment			T7651 As delivered										
8	Sample Test piece Heat treatment			T7651										
9	Dimension concerned	a	mm	$120 < a \leq 160$			$160 < a \leq 180$			$180 < a \leq 200$				
10	Thickness of cladding on each face		%	-										
11	Direction of test piece			L	LT	ST	L	LT	ST	L	LT	ST		
12	TENSILE	Temperature	θ	°C	Room temperature									
13		Proof stress	R _{p0,2}	MPa	430	425	400	425	420	400	415	415	390	
14		Strength	R _m	MPa	490	495	470	485	490	470	480	485	460	
15		Elongation	A _{5D}	%	5	4	3	4	3	3	4	3	3	
16		Reduction of area	Z	%	-	-	-	-	-	-	-	-	-	
17	Hardness		HB	-	-									
18	Shear strength		R _c	MPa	-									
19	Bending		k	-	-									
20	Impact strength		k	-	-									
21	CREEP	Temperature	θ	°C	-									
22		Time		h	-									
23		Stress	σ_a	MPa	-									
24		Elongation	a	%	-									
25		Rupture stress	σ_R	MPa	-									
26		Elongation at rupture	A	%	-									
27	Notes (see Line 98)			1), 2), 3), and 4)										

32	Electrical conductivity	7	γ (MS/m)	$R_{p0.2}$		Action	
			$\gamma \geq 22.5$	$R_{p0.2LT} \leq R_{p0.2LTmin} + 70MPa$		Acceptable	
			$\gamma \geq 22.5$	$R_{p0.2LT} > R_{p0.2LTmin} + 70MPa$		Additional artificial age at: 169 < θ < 175°C (3) (4)	
			$\gamma < 22.5$	Not Acceptable			
39	Stress corrosion	3	a > 25mm				
		6	ST direction, Constant Load, Stress level 170 MPa				
		7	Criteria: no failure 30 days				
40	Fracture toughness (K_{1C})	3	CT specimen				
		7	Dimension, mm	L-T MPa√m	T-L MPa√m	S-L MPa√m	
			25 < a ≤ 40	28	26	-	
			40 < a ≤ 50	27	25	-	
			50 < a ≤ 60	27	25	25	
			60 < a ≤ 80	26	25	25	
			80 < a ≤ 160	25	24	24	
			160 < a ≤ 200	24	23	23	
40	R-curve (K_{c0})	3	CCT specimen, W = 406mm, 2ao/W = 0,25 Thickness = 6,3mm L-T direction				
		7	The toughness shall not be lower than the following data:				
			6,0 ≤ a ≤ 60,0	a/2	80		
			60,0 < a ≤ 100,0	a/4	60		
			100,0 < a ≤ 150,0	a/4	50		
			150,0 < a ≤ 200,0	a/4	40		
44	External defects	-	See AIMS 03-02-000				
46	Fatigue	3	T type specimen in accordance with EN6072, 25 ≤ a ≤ 150				
		7	The curve of the test results, calculated in the accordance with EN6072 shall not lie below the reference curve defined by the following data				
			$\sigma_{max\ nett}$ (MPa)	No. of cycles (N)	R	Kt	Direction
			280	1,0E4	0,1	2,3	L-T & T-L
			200	3,0E4			
			160	1,0E5			
			150	2,0E5			
			130	1,0E6			

46	Fatigue	3	T type specimen in accordance with EN6072, 150 < a ≤ 200				
		7	The curve of the test results, calculated in the accordance with EN6072 shall not lie below the reference curve defined by the following data				
			σ _{max nett} (MPa)	No. of cycles (N)	R	Kt	Direction
			270	1,0E4	0,1	2,3	L-T & T-L
			190	3,0E4			
			150	1,0E5			
			140	2,0E5			
			120	1,0E6			
46	Fatigue (Smooth)	-	See AIMS 03-02-000				
		6	Thickness: 150 ≤ a ≤ 200				
		7	The specimen shall be tested at R=0,1 at a maximum stress of 240MPa and shall meet the following fatigue requirements:				
			Minimum Individual: 90000 cycles Log average of 4 samples: 120000 cycles Runout: 200000 cycles				
49	Exfoliation corrosion	3	a/2 position				
		7	Criteria: Better or equal to grade EB of ASTM G34				
61	Internal defects	-	See AIMS 03-02-000				
69	Elastic modulus	3	L Direction				
		7	The value must be within the following range: 70000 ≤ E ≤ 74000 MPa 71000 ≤ E _c ≤ 75000 MPa				
70	Compression Proof stress R _{0,2C}	7	Dimension, mm	L MPa	LT MPa	ST MPa	
			6 ≤ a ≤ 60	450	465	-	
			60 < a ≤ 100	440	455	-	
			100 < a ≤ 160	415	435	430	
			160 < a ≤ 200	410	430	425	

71	Crack propagation	1	ASTM E647						
		3	CCT Specimen, W = 160mm, 2a _i = 4,0mm, 2a _f = 53,0mm, thickness = 5,0mm, L-T and T-L						
		7	The crack propagation rates shall not be greater than the following data:						
			ΔK (MPa√m)		da/dN (mm/cycle)		R		
			10		1,5E-04		0,1		
			15		5,0E-04				
			20		1,5E-03				
			25		5,0E-03				
			30		1,0E-02				
82	Batch uniformity	1	See AIMS 03-02-000						
		7	Electrical Conductivity	γ = 23,4MS/m (Typical Value)					
δ ≤ 1,0 MS/m per product				Δ ≤ 1,5 MS/m per batch					
85	Bearing stress	7	Dimensions mm		25 < a ≤ 50	50 < a ≤ 75	75 < a ≤ 100	100 < a ≤ 200	
					Fbru	MPa	e/D= 2,0	980	1000
					e/D= 1,5	760	780	760	740
			Fbry	MPa	e/D= 2,0	730	750	710	700
					e/D= 1,5	630	640	630	610
98	Notes	-	1) See EN 2032-2 2) Where gauge length, A = 50mm 3) The certificate of conformity shall be specially endorsed and the recommended heat treatment specifically stated. 4) Materials shall be considered to have acceptable stress corrosion resistance provided that upon completion of this additional ageing it meets the requirements of lines 13, 14, 15 and the acceptance criteria of line 32.						
99	Typical use	-							
100	Qualification								

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 03/10		New standard
2 04/11	Line 40 Line 46	Change to R-Curve requirements Removal of S-L fatigue requirement from specification Change to Specimen Type. Replacement of Mini T Type with T type specimen
3 09/11	Line 13 & 14	Improve spec min for thickness ranges $160 < a \leq 180$ & $180 < a \leq 200$